End-to-End DevOps CI/CD Pipeline on AWS

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Herovire Capstone, Batch 15

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Agenda

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2. Architecture
3. Build and deploy pipeline (Jenkins)
4. Infrastructure as code (Terraform)
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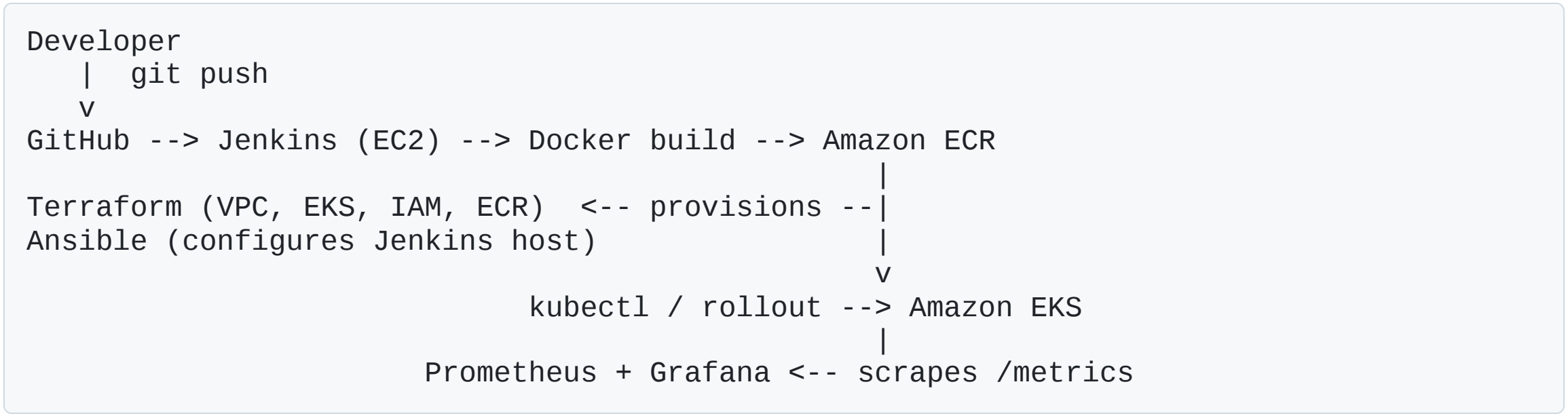
What I built

A pipeline that takes my app from a `git push` to a running, monitored service on AWS, without any manual steps.

What it had to do:

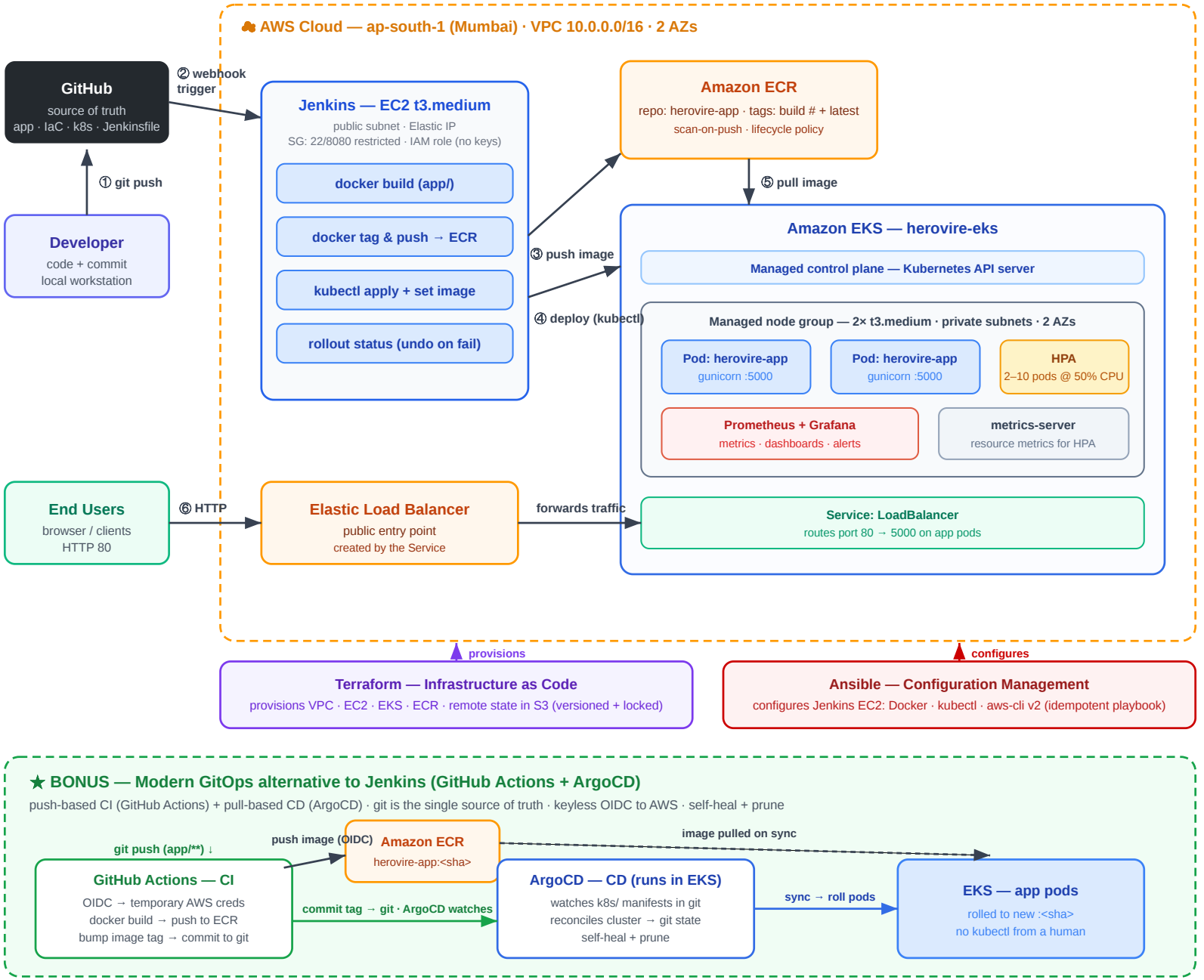
- Containerize the app and build it automatically
- Create all AWS infrastructure from code, not the console
- Deploy to a Kubernetes cluster
- Monitor it and alert when something breaks
- Stay cheap and tear down cleanly

Architecture



Herovire DevOps Capstone — End-to-End CI/CD Architecture on AWS

GitHub → Jenkins → Docker → ECR → EKS · provisioned with Terraform · configured with Ansible · monitored with Prometheus + Grafana



Network layout

```
VPC 10.0.0.0/16  (ap-south-1, two AZs)
│
├── Public  10.0.1.0/24, 10.0.2.0/24
│    ├── Jenkins EC2 (t3.medium, static EIP)
│    └── Load balancer for the app
│
├── Private 10.0.11.0/24, 10.0.12.0/24
│    └── EKS worker nodes, no public IPs
│
├── Internet Gateway   (public subnets)
└── NAT Gateway (one)  (private subnets reach out)
```

Nodes are private, so the only way in is through the load balancer.

One NAT gateway instead of one per AZ: a single point of failure,

but it halves the NAT cost and is the right trade here.

Tools I used

Stage	Tool	What it does
Build	Docker + Jenkins	Build the image, push to ECR
Infrastructure	Terraform	VPC, EKS, ECR, IAM, state in S3
Configuration	Ansible	Sets up the Jenkins server
Deploy	kubectl and EKS	Rolls the image onto the cluster
Monitoring	Prometheus + Grafana	Metrics, dashboard, alerts

The app itself is a small Flask service with `/`, `/health` and `/metrics`.

Build and deploy (Jenkins)

The `Jenkinsfile` pipeline runs:

```
Checkout -> Build -> ECR login -> Tag and Push -> Deploy -> Smoke Test
```

- Images go to ECR, tagged with the build number
- Deploy runs `kubectl apply` then updates the image tag
- `kubectl rollout status` gates the build, so it only passes if pods come up healthy
- If the smoke test fails, the pipeline rolls back with `kubectl rollout undo`

Infrastructure as code (Terraform)

Everything on AWS is defined in `terraform/`, split into two root modules with separate state:

Module	Contains	Applied by
`bootstrap/`	VPC, subnets, NAT, Jenkins EC2	me, from a laptop
`platform/`	EKS, ECR, OIDC role	the Jenkins pipeline

State lives in a versioned S3 bucket, so the environment is reproducible. `terraform plan` costs nothing, so I only ran `apply` when I was actually working.

Configuration (Ansible)

Two layers on the Jenkins host:

- **EC2 user data** bootstraps it at first boot: Java 21, Jenkins, Docker, Terraform
- **Ansible** (`configure-nodes.yml`) then installs Docker, kubectl and the AWS CLI, and points kubeconfig at the EKS cluster

Ansible is idempotent, so re-running it converges to the same state. User data only ever runs once, on first boot, which is why the ongoing config lives in Ansible.

Kubernetes

- Deployment with 2 replicas
- Service of type LoadBalancer, so the app is reachable publicly
- HPA to scale on CPU, 2 to 10 pods (needs metrics-server installed)
- Liveness and readiness probes on `/health`

Monitoring

- Installed kube-prometheus-stack with Helm (Prometheus, Grafana, Alertmanager)
- The app uses `prometheus-flask-exporter` to expose `/metrics`
- A ServiceMonitor tells Prometheus to scrape it
- Grafana dashboard shows request rate: `rate(flask_http_request_total[1m])`
- Alert `AppPodDown` fires when the scrape target stops responding

Jenkins is monitored too, not just the app. If the CI server is down, nothing can build or deploy, so it belongs on the same dashboards.

Monitoring the CI server

Jenkins runs on EC2, outside the cluster, so there is no Service for a ServiceMonitor to select. It is scraped as a static target instead.

- Prometheus metrics plugin exposes `/prometheus/` on the Jenkins host
- Prometheus authenticates with an API token mounted from a Kubernetes secret, so no token is committed
- Grafana dashboard "Jenkins CI": build queue, executor usage, node health
- Alerts `JenkinsDown` and `JenkinsBuildQueueStuck`

The security group had to allow 8080 from inside the VPC, because the nodes scrape from the private subnets.

Security

No static AWS credentials anywhere in the pipeline.

Identity	Permissions
Jenkins EC2 role	`AdministratorAccess`, broad on purpose
EKS node role	Worker, CNI, and ECR read only
GitHub Actions	ECR power user, assumed via OIDC

- Jenkins authenticates to AWS with an instance role, not access keys
- Worker nodes are private, with no public IPs
- Ingress is restricted: 22 and 8080 from my IP, 8080 from GitHub's webhook ranges, and 22 within the group itself for the Ansible stage

The Jenkins role is admin because it runs `terraform apply` across VPC, IAM, EKS and EC2. In production this would be a scoped provisioning role. I have called it out in the code rather than hidden it.

Evidence it runs

Both Jenkins pipelines went green as real jobs, not from my laptop:

Pipeline	Result
`herovire-app`	6 stages green, image pushed, smoke test passed
`herovire-infra`	18 resources added, Ansible `ok=8 changed=3`, 2 nodes Ready

- I applied `bootstrap` by hand; **Jenkins applied `platform`**, creating the cluster, node group and ECR from an empty state
- Prometheus then showed both targets up, `jenkins` and `herovire-app`
- Teardown removed all 48 resources and returned the account to zero

Screenshots of the green builds, the Prometheus targets and the Grafana dashboards are in `docs/screenshots/`.

Testing

- `tests/smoke_test.sh` checks that `/` and `/health` return 200
- `tests/infra_test.sh` checks nodes are Ready, pods Running, LB is up
- The smoke test runs as a Jenkins stage, so a bad deploy fails the build
- I ran the full cycle end to end: apply, test, destroy

Bonus: GitOps with GitHub Actions and ArgoCD

I also built a second, more modern path and ran it live:

- GitHub Actions builds and pushes the image, authenticating with OIDC so no AWS keys are stored in GitHub
- It then updates the image tag in `k8s/deployment.yaml` and commits it
- ArgoCD watches the repo and syncs the cluster to match

One push took the app from v1 to v2 with no `kubectl` from me.

Demo

1. App running as v1 on its LoadBalancer URL
2. Push a small change
3. GitHub Actions goes green
4. ArgoCD flips from OutOfSync to Synced, pods roll
5. Refresh, app is now v2
6. `kubectl scale --replicas=1`, ArgoCD puts it back to 2

Problems I ran into

- My Mac builds arm64 images but the nodes are amd64, so pods failed to pull. Fixed by building with `--platform linux/amd64`
- Jenkins was installed but dead on every fresh deploy, because a failing `pip install` in user data stopped the script before Jenkins started
- Jenkins could not reach the cluster: it sits inside the VPC, so the EKS endpoint resolved to private IPs the security group did not allow
- Leftover LoadBalancers blocked the VPC from deleting, so I now delete the Services before running destroy
- Prometheus was scraping nothing because the Service was missing an `app` label

Cost

- Running cost is roughly $0.25 to $0.30 per hour (EKS, 2 nodes, NAT, LB)
- Writing code and running `plan` is free, only `apply` costs anything
- Kept it small: 2 t3.medium nodes, 3 day metrics retention, one NAT gateway
- Destroyed everything after each session, so a full build and test cycle cost about $0.25
- ECR lifecycle policy clears out untagged images

What I would do next

- Scope the Jenkins IAM role down from `AdministratorAccess`
- Move the infra pipeline onto ephemeral runners so nothing long lived needs admin rights
- Add a Cluster Autoscaler, since today pods would sit `Pending` past 3 nodes
- Put the app behind an ALB with TLS instead of a Classic Load Balancer
- Rewrite `AppPodDown` as an `absent()` rule so it catches a true zero-pod state

What I learned

- Infrastructure as code makes environments reproducible and disposable
- Terraform and Ansible solve different problems
- Push based delivery (Jenkins) and pull based delivery (ArgoCD) are both valid
- OIDC and short lived credentials are better than storing secrets
- A deploy is not finished until you can see that it is healthy
- Cost control has to be a habit, not something you fix at the end

Summary

GitHub to Jenkins to Docker to ECR to Terraform to Ansible to EKS,
with Prometheus and Grafana on top, plus a working GitOps path.

One `git push` gets me a running, monitored app on Kubernetes.

Thank you. Questions?